

Introductory
Night Sky
photography

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Time-lapse Photography



The background of the entire image is a twilight sky, transitioning from a deep purple at the top to a warm orange near the horizon. A series of approximately 25 crescent moon images are arranged in a smooth, downward-curving arc from the upper left towards the lower right. The moons vary slightly in phase and size, suggesting a time-lapse or stacked image sequence. In the bottom left, the dark, silhouetted branches of trees are visible against the sky. In the bottom right, some distant city lights and more tree silhouettes are faintly visible.

Movement

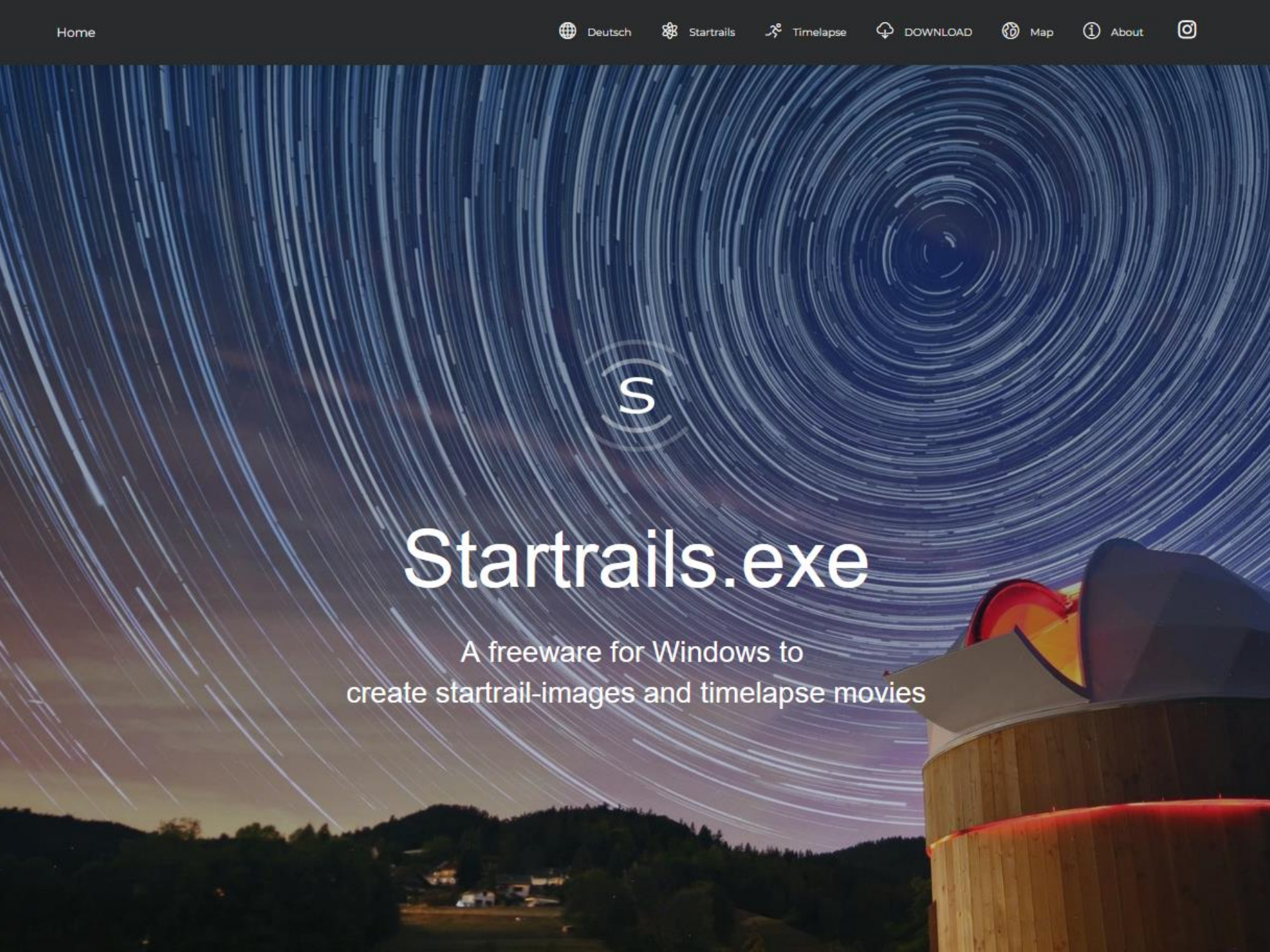
Moon trails

Take a series of photographs
to stack or turn into a video.



Startrails.exe

A freeware for Windows to
create startrail-images and timelapse movies



File Build Options ?



Donate

Images



Files:

- ☒ IMG0001.jpg
- ☒ IMG0002.jpg
- ☒ IMG0003.jpg
- ☒ IMG0004.jpg
- ☒ IMG0005.jpg
- ☒ IMG0006.jpg
- ☒ IMG0007.jpg
- ☒ IMG0008.jpg
- ☒ IMG0009.jpg
- ☒ IMG0010.jpg
- ☒ IMG0011.jpg
- ☒ IMG0012.jpg
- ☒ IMG0013.jpg
- ☒ IMG0014.jpg
- ☒ IMG0015.jpg

Darkframes



Files

Selection

Foreground (averaged)

Result



Movement

Moon trails

Stacked using startrails.de

Exposures:

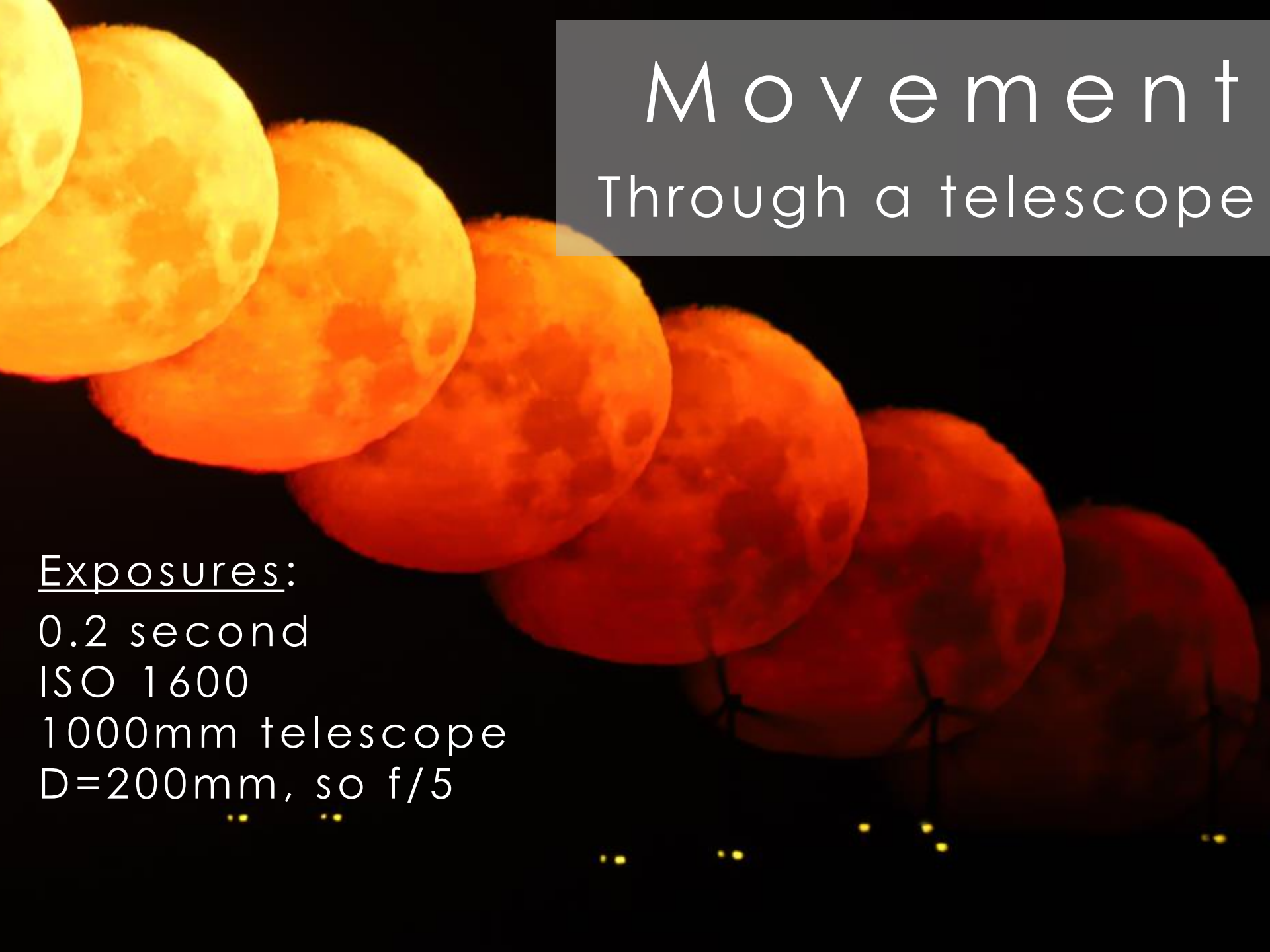
1 second

ISO 1000

f/8

300mm lens





Movement Through a telescope

Exposures:

0.2 second

ISO 1600

1000mm telescope

$D=200\text{mm}$, so $f/5$

Planning with Stellarium and a map

Library Square

University of Sussex
Students' Union (USSU)

The Bridge Café

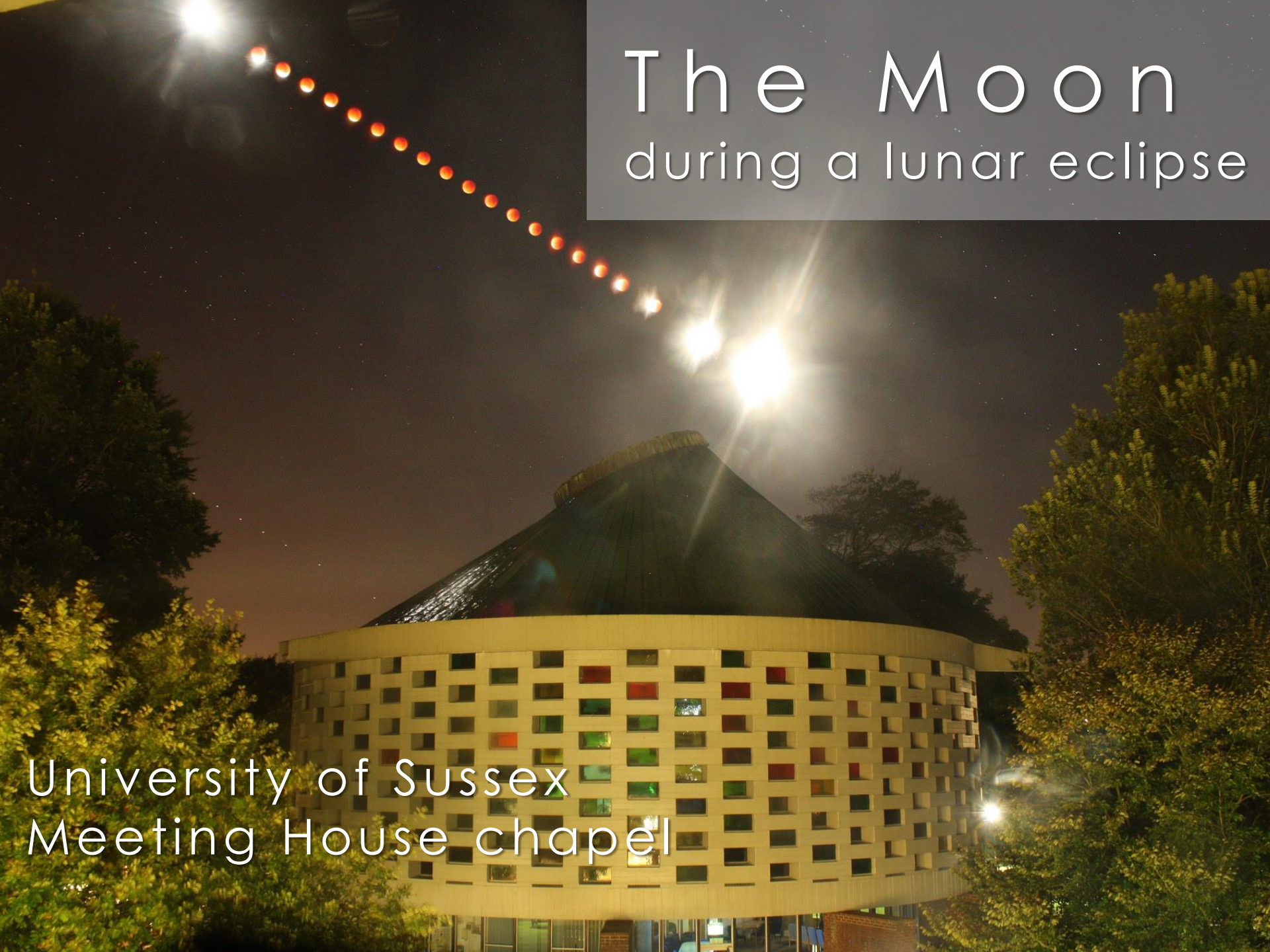
N-S Rd

N-S Rd

The Moon

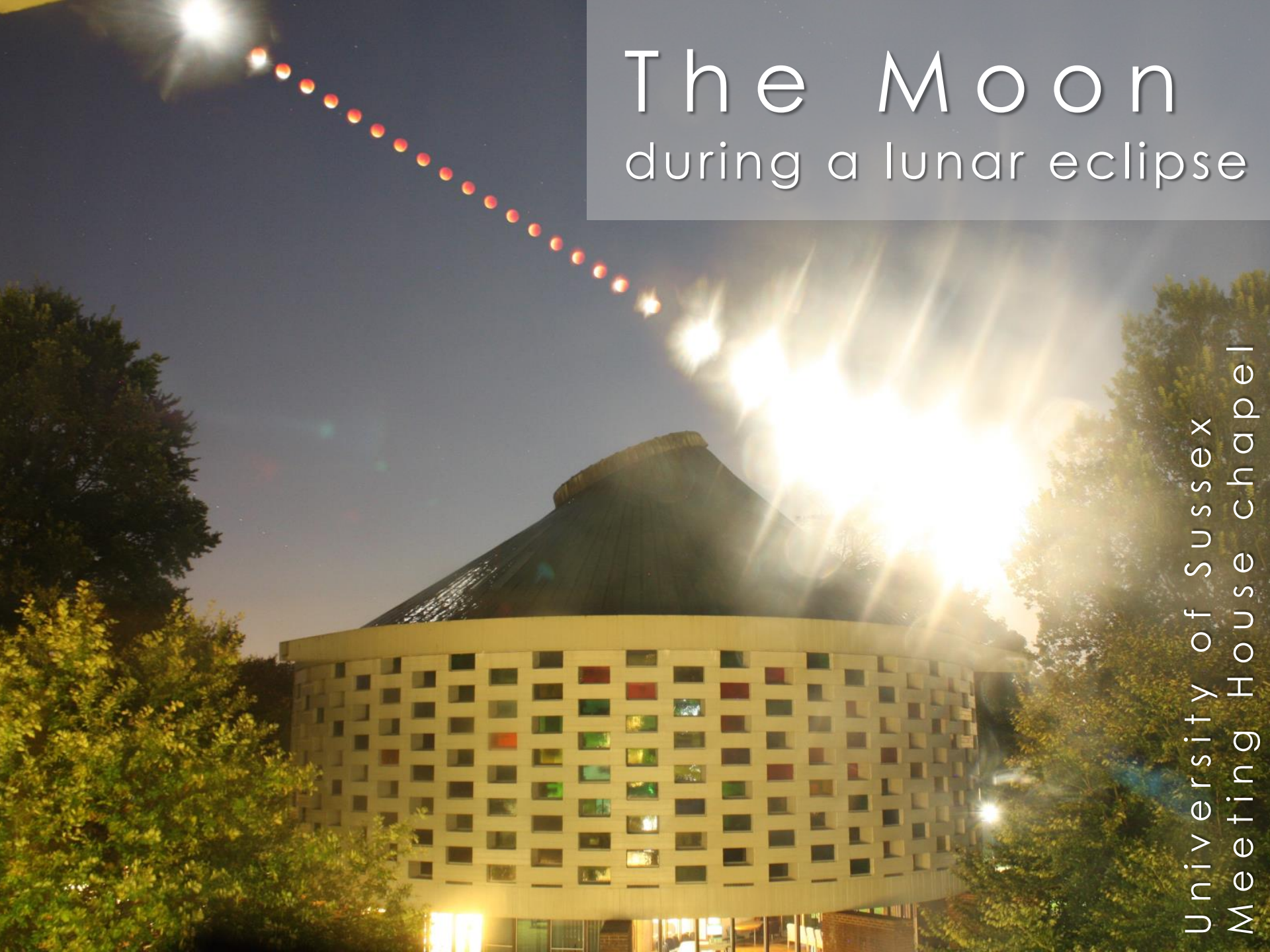
during a lunar eclipse

University of Sussex
Meeting House chapel



The Moon

during a lunar eclipse



University of Sussex
Meeting House chapel

The Moon

during a lunar eclipse



University of Sussex
Meeting House chapel



Next lunar eclipse

Friday 28th August

02:23 – (05:12) – 06:17



A long-exposure photograph of a night sky showing numerous concentric, curved star trails. The trails are most prominent in the upper left and center, where they form dense, circular patterns. The lower portion of the image is filled with the dark, silhouetted branches and foliage of trees, which appear to be in the foreground. The overall scene is a serene and awe-inspiring depiction of celestial motion.

Movement

Star trails

Movement

Star trails





Exposures:

30 seconds

f/9

ISO 3200

Repeated for 8 hours

At f/9, both badger
& stars are in focus

Camera lens warmer

Prevents fog and dew from forming in cold weather and extremely cold areas.



Without Lens Heater



With Lens Heater

Essential if your camera will be outside for long periods of time taking a time-lapse



Movement

Star trails



Mauna Kea Hawai'i



~50 images, stacked (using the Startrails software)

Mauna Kea Hawaii'i

~30 images, aligned and stacked (using Sequator)

Mauna Kea Hawai'i



~50 images, aligned and stacked
(after the Moon has risen to illuminate the scene)

Mauna Kea Hawai'i



The stacked image PLUS the aligned & stacked image
(to get both star trails and the Milky Way superimposed)

Mauna Kea Hawai'i



The stacked image PLUS the aligned & stacked image
(to get both star trails and the Milky Way superimposed)

Astrophotography

Any Questions?



Astronomy Festival Programme

Friday 28th - Sunday 30th
August 2026



Astronomy Festival Programme

Friday 28th - Sunday 30th August 2026

First time at the Observatory Science Centre & Museum? We have over 100 interactive science exhibits to explore!

Themed indoor galleries - 'Earth & Beyond', 'Light & Colour' and 'Forces', 'Puzzles & Illusions' and 'Ancient Wisdom'

Outdoor Discovery Park - our 'science playground' of large-scale interactive exhibits

Outdoor water play areas - Build dams, explore water pressure, and create locks and canals.

Historic Telescopes - Each dome contains an historic telescope.

The telescopes in domes B, E can be visited during your daytime visit.

Telescopes in domes A, C and D can be visited during our evening sessions.

The Lunar Experience - Dome F is currently the home of 'Lunar', a walkthrough science show where you will uncover the mysteries of the moon. Lunar is available during daytime sessions, pick up your free tickets at reception.

Telescope tours - Find out more about the telescopes during our free telescope tours.

The Cafe - Open for light meals, snacks and hot/cold drinks during each session.

There's always more to explore...

Become a member!



Your £45 Annual membership includes:

12 months' unlimited daytime entry to the Observatory Science Centre & Museum

Valid weekends and East Sussex school holidays, during public opening hours

A unique QR code delivered by email for quick scanning on arrival at reception.

Purchase your annual membership online or sign up at reception.



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<http://www.DarrenBaskill.co.uk>

